

**AGNELS CHARITIES FR. C. RODRIGUES
INSTITUTE OF TECHNOLOGY**

Vashi, Navi Mumbai

**HOSPITALITY REVENUE
ANALYTICS**

**A Data Analytics Project Using Python, SQL and
Tableau**

Project Report

Student: Kartik Solunke

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Abstract

The hospitality industry generates large volumes of booking, customer, pricing and cancellation data. Converting this data into actionable business information can help management understand revenue performance, customer behaviour and booking patterns. This project presents a hospitality revenue analytics solution developed using Python, SQL and Tableau.

The dataset contains 87,366 booking records and 45 attributes covering hotel type, lead time, stay duration, customer segment, booking platform, payment method, ADR, discounts, booking status and revenue. Python and Pandas were used for data inspection, validation and feature-level analysis. SQL was used to perform structured business analysis such as hotel performance, customer segmentation, lead-time analysis, repeat-guest analysis, stay-duration analysis and discount impact analysis. Tableau was used to transform the results into interactive dashboards and KPI-based visualizations.

The analysis identified several important patterns. Total final revenue was approximately INR 31.01 million from 87,366 bookings. Business customers contributed approximately INR 18.64 million, while the mobile app generated approximately INR 15.40 million in final revenue. Cancellation rates increased with longer booking lead times, reaching approximately 40% for bookings made more than 180 days in advance. Repeat guests had a cancellation rate of approximately 15.38%, compared with approximately 28.73% for non-repeat guests. Increasing discounts from 0% to 20% were associated with progressively lower final revenue in the dataset, while cancellation rates remained close to 28%.

Keywords: Hospitality Analytics, Revenue Analytics, Python, Pandas, SQL, Tableau, Customer Segmentation, Cancellation Analysis, Business Intelligence

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Chapter 1

Introduction

1.1 Background

The hospitality sector depends heavily on bookings, room pricing, customer behaviour and distribution channels. Hotels need to understand not only how many bookings they receive, but also how those bookings translate into revenue, how frequently they are cancelled and which customer and digital channels contribute most to business performance.

Data analytics provides a systematic way to convert historical booking records into measurable business insights. By combining data preparation, SQL-based analysis and dashboard visualization, hotel management can monitor key performance indicators and identify patterns that may support operational and commercial decisions.

1.2 Problem Statement

Hospitality businesses generate large volumes of booking and customer data, but raw transactional data alone does not provide clear visibility into revenue performance, customer behavior, booking patterns, and cancellation trends. Identifying the factors that influence these business outcomes is essential for making informed, data-driven decisions.

This project addresses this challenge by transforming raw hospitality booking data into actionable business insights. The analysis integrates Python-based data preparation and exploratory analysis with SQL-based business analysis and interactive Tableau dashboards. Key performance indicators such as revenue, bookings, ADR, cancellation rate, customer segments, booking platforms, lead time, stay duration, and discount impact are analyzed to identify meaningful patterns and performance drivers.

The final outcome is an end-to-end analytics solution that converts raw booking data

into decision-ready insights, enabling stakeholders to monitor performance, understand customer behavior, and identify areas for revenue and operational improvement.

1.3 Objectives

1. Inspect and validate the hospitality booking dataset.
2. Identify missing values, duplicate records and inconsistent data patterns.
3. Perform exploratory data analysis using Python and Pandas.
4. Create and validate revenue-related analytical fields.
5. Use SQL to analyse revenue, bookings, ADR and cancellation behaviour.
6. Compare hotel types, customer segments, booking platforms and payment methods.
7. Analyse the relationship of lead time, stay duration and discounts with business performance.
8. Develop interactive Tableau dashboards containing KPIs and visualizations.
9. Summarize the major findings and provide data-supported business observations.

1.4 Scope

The project focuses on historical booking-level analytics. The analysis covers revenue performance, booking volume, ADR, cancellation rate, customer segments, booking platforms, payment methods, lead time, repeat guests, stay duration and discount levels. The project is intended for descriptive and diagnostic analytics rather than real-time hotel revenue management or predictive forecasting.

Chapter 2

Dataset and Data Description

2.1 Dataset Overview

The final dataset used for the analysis contains **87,366 booking records** and **45 variables**, providing a substantial foundation for business-oriented hospitality analytics. Each row represents a booking-level record, enabling detailed analysis of customer behavior, hotel performance, booking patterns, pricing, and cancellations.

The dataset covers multiple analytical dimensions, including hotel and customer information, booking characteristics, stay duration, pricing and discount information, booking status, payment methods, booking platforms, and derived revenue metrics. This multidimensional structure allows the project to move beyond descriptive reporting and identify patterns and relationships that can support data-driven business decision-making.

2.2 Dataset Characteristics

The final dataset contains 87,366 booking records and 45 variables. Each record represents an individual hotel booking, covering hotel type, customer information, booking behaviour, pricing, stay duration, payment methods, cancellations and revenue.

The dataset includes both categorical and numerical variables, providing multiple dimensions for analysing revenue performance, customer behaviour and booking patterns. It serves as the common data source for Python-based validation, SQL analysis and Tableau visualization.

2.3 Major Variables

Table 2.1: Major Variables

Variable	Description
HOTEL	Hotel type/category associated with the booking.
IS_CANCELED	Cancellation indicator.
LEAD_TIME	Number of days between booking and arrival.
STAYS_IN_WEEKEND_NIGHTS	Number of weekend nights.
STAYS_IN_WEEK_NIGHTS	Number of weekday nights.
TOTAL_NIGHTS	Total length of stay.
ADULTS, CHILDREN, BABIES	Number of guests by category.
CUSTOMER_SEGMENT	Business, Individual or Group customer classification.
BOOKING_PLATFORM	Mobile App, Website or Partner Platform.
PAYMENT_METHOD	Payment method used for the booking.
ADR	Average Daily Rate associated with the booking.
DISCOUNT_PERCENTAGE	Discount percentage applied to the booking.
BOOKING_STATUS	Completed or cancelled booking status.
REVENUE	Revenue field available in the dataset.
TOTAL_REVENUE	Total revenue field available in the dataset.
FINAL_REVENUE	Final revenue field used for the main revenue analysis.

2.4 Dataset Validation

Several SQL validation checks were performed before the business analysis.

- Total records checked: **87,366**.
- Duplicate-row check: no duplicate rows were identified in the final CSV inspection.
- Selected null-value checks for HOTEL, LEAD_TIME, ARRIVAL_DATE, ADULTS, CHILDREN, ADR and REVENUE returned zero null values.
- TOTAL_NIGHTS was checked against STAYS_IN_WEEKEND_NIGHTS + STAYS_IN_WEEK_NIGHTS, the validation returned **0 incorrect records**.

- FINAL REVENUE was checked against the discount formula used in the dataset and returned **0 inconsistent records** under the selected tolerance.
- ADR was inspected for zero values because zero ADR bookings can represent bookings that did not generate room revenue.

2.5 Revenue Calculation

The final revenue field was validated using the following relationship:

$$\text{Final Revenue} = \text{Revenue} - \left(\text{Revenue} \times \frac{\text{Discount Percentage}}{100} \right) \quad (2.1)$$

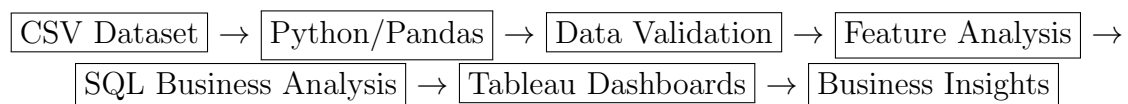
The validation query confirmed that the stored FINAL REVENUE values were consistent with this formula within the selected numerical tolerance.

Chapter 3

Methodology

3.1 Project Workflow

The overall workflow followed a structured data analytics pipeline:



3.2 Python and Pandas

Python was used for data inspection, data quality checks, exploratory analysis and preparation of analytical fields. Pandas provided DataFrame-based operations for filtering, aggregation and validation, while numerical and visualization libraries supported the exploratory stage.

3.3 SQL Analysis

MySQL was used to perform structured business analysis. SQL aggregation functions such as `COUNT()`, `SUM()` and `AVG()` were combined with `GROUP BY`, `ORDER BY`, `WHERE` and conditional expressions.

A representative query used for overall performance was:

```
SELECT
    COUNT(*) AS total_bookings,
    SUM(FINAL_REVENUE) AS total_revenue,
    AVG(ADR) AS average_adr,
```

```
SUM(BOOKING_STATUS = 'cancelled') * 100.0 / COUNT(*)  
    AS cancellation_rate  
FROM hospitality_revenue_sql;
```

3.4 Tableau Visualization

Tableau was used to create interactive dashboards containing KPI cards, revenue trends, hotel comparisons, customer-segment analysis, booking-platform analysis and cancellation-related views. Filters and dashboard interactions were used to support exploration of the data.

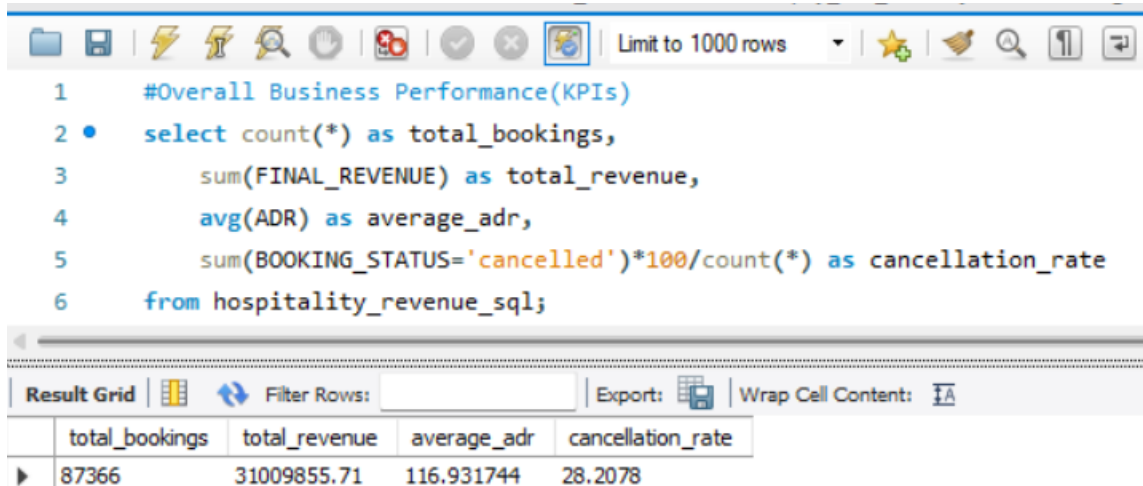
Chapter 4

SQL Analysis

4.1 Overall Business Performance

SQL was used to analyze the hospitality dataset and derive business-oriented insights. The analysis was performed using MySQL queries covering overall performance, hotel performance, customer segments, booking platforms, lead time, stay duration, and discount impact.

The following SQL query calculates total bookings, total revenue, average ADR, and cancellation rate.



The screenshot shows a MySQL query editor interface. The top toolbar includes icons for file operations, execution, and settings. The query text is as follows:

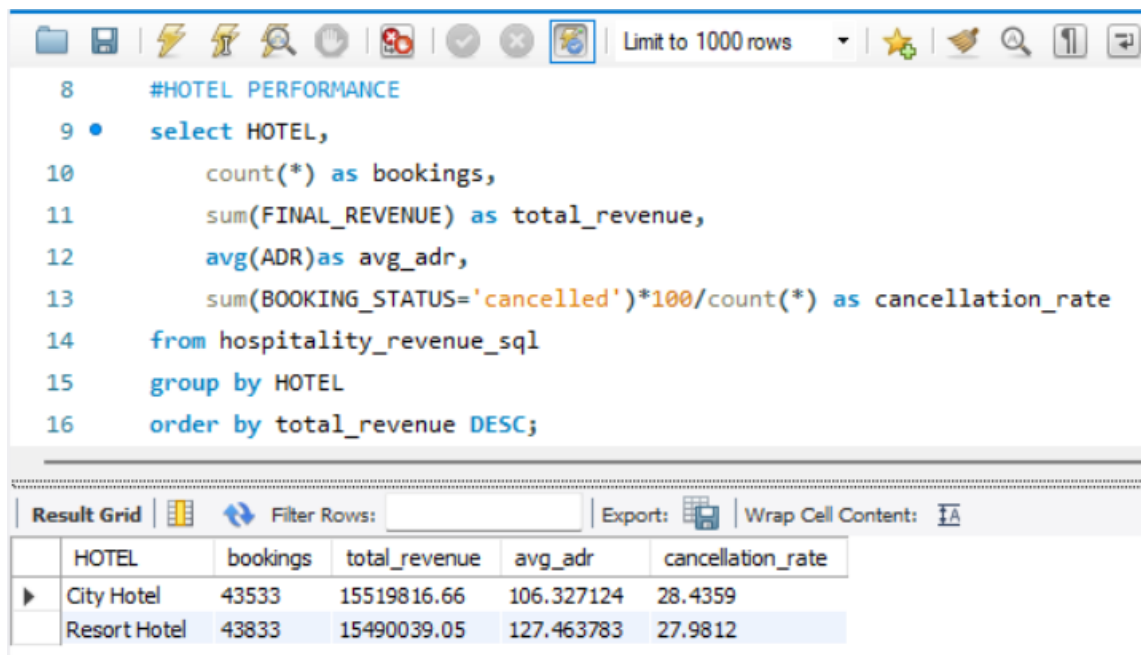
```
1  #Overall Business Performance(KPIs)
2  •  select count(*) as total_bookings,
3      sum(FINAL_REVENUE) as total_revenue,
4      avg(ADR) as average_adr,
5      sum(BOOKING_STATUS='cancelled')*100/count(*) as cancellation_rate
6  from hospitality_revenue_sql;
```

Below the query editor, the results are displayed in a table with the following columns and values:

	total_bookings	total_revenue	average_adr	cancellation_rate
▶	87366	31009855.71	116.931744	28.2078

Figure 4.1: SQL Result: Overall Business Performance

4.2 Hotel Performance



The screenshot shows a SQL query editor with a toolbar at the top. The query is as follows:

```
8 #HOTEL PERFORMANCE
9 • select HOTEL,
10     count(*) as bookings,
11     sum(FINAL_REVENUE) as total_revenue,
12     avg(ADR) as avg_adr,
13     sum(BOOKING_STATUS='cancelled')*100/count(*) as cancellation_rate
14 from hospitality_revenue_sql
15 group by HOTEL
16 order by total_revenue DESC;
```

Below the query, the results are displayed in a table with the following columns: HOTEL, bookings, total_revenue, avg_adr, and cancellation_rate. The results are ordered by total_revenue in descending order.

HOTEL	bookings	total_revenue	avg_adr	cancellation_rate
City Hotel	43533	15519816.66	106.327124	28.4359
Resort Hotel	43833	15490039.05	127.463783	27.9812

Figure 4.2: SQL Result: Hotel Performance

The two hotel categories have very similar booking volumes and total final revenue. The Resort Hotel records a higher average ADR, while the City Hotel has a marginally higher total final revenue in the SQL output.

4.3 Customer Segment Analysis

```
18  #CUSTOMER SEGMENT
19  •  select CUSTOMER_SEGMENT,
20         count(*) as bookings,
21         sum(FINAL_REVENUE) as total_revenue,
22         avg(ADR) as avg_adr,
23         sum(BOOKING_STATUS='cancelled')*100/count(*) as cancellation_rate
24  from hospitality_revenue_sql
25  group by CUSTOMER_SEGMENT
26  order by total_revenue desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	CUSTOMER_SEGMENT	bookings	total_revenue	avg_adr	cancellation_rate
▶	Business	52333	18640679.51	117.218959	28.1925
	Individual	26248	9245594.16	116.339899	28.0745
	Group	8785	3123582.04	116.989101	28.6966

Figure 4.3: SQL Result: Customer Segment Analysis

The Business segment is the largest segment by booking volume and final revenue in the dataset.

4.4 Booking Platform Analysis

```
28      #BOOKING PLATFORM PERFORMANCE
29 •    select BOOKING_PLATFORM, count(*) as bookings,
30           sum(FINAL_REVENUE) as total_revenue,
31           avg(ADR) as avg_adr,
32           sum(BOOKING_STATUS='cancelled')*100/count(*)as cancellatio_rate
33 from hospitality_revenue_sql
34 group by BOOKING_PLATFORM
35 order by total_revenue desc;
--
```

	BOOKING_PLATFORM	bookings	total_revenue	avg_adr	cancellatio_rate
▶	Mobile App	43508	15403460.52	117.136335	28.0201
	Website	25497	9079687.66	116.790049	28.3563
	Partner Platform	18361	6526707.53	116.643709	28.4462

Figure 4.4: SQL Result: Booking Platform Performance

The Mobile App has the largest booking volume and the highest final revenue among the three booking platforms.

4.5 Payment Method Analysis

The payment-method analysis showed the following final-revenue values:

```
37 #PAYMENT METHOD PERFORMANCE
38 • select Payment_Method, count(*) as bookings,
39        sum(FINAL_REVENUE) as total_revenue,
40        avg(ADR) as avg_adr,
41        sum(BOOKING_STATUS='cancelled')*100/count(*)as cancellatio_rate
42 from hospitality_revenue_sql
43 group by Payment_Method
44 order by total_revenue desc;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 					
	Payment_Method	bookings	total_revenue	avg_adr	cancellatio_rate
▶	UPI	25435	8982634.62	116.613512	28.2052
	DEBIT CARD	22721	8155719.93	117.149959	28.4979
	NET BANKING	20794	7377608.35	116.828103	28.4313
	CASH	18416	6493892.81	117.219062	27.6010

Figure 4.5: SQL Result: Payment Method Analysis

4.6 Lead Time Analysis

```

55     #LEAD TIME ANALYSIS
56 •   SELECT
57     case when LEAD_TIME<=7 then '0 - 7 days'
58         when LEAD_TIME<=30 then '8 - 30 days'
59         when LEAD_TIME<=90 then '31 - 90 days'
60         when LEAD_TIME<=180 then '91 - 180 days'
61         else '180+ days'
62     end as lead_time_group,
63     count(*) as bookings,
64     sum(FINAL_REVENUE) as total_revenue,
65     avg(ADR) as avg_adr,
66     sum(BOOKING_STATUS='cancelled')*100/count(*) as cancellation_rate
67 from hospitality_revenue_sql
68 group by lead_time_group
69 order by bookings desc;

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 					
lead_time_group	bookings	total_revenue	avg_adr	cancellation_rate	
31 - 90 days	22740	8614315.69	122.077992	32.3747	
0 - 7 days	18287	3181026.53	100.471901	10.5594	
91 - 180 days	18243	8486836.93	125.524562	35.1642	
8 - 30 days	16332	5189004.01	121.653867	125.524562	
180+ days	11764	5538672.55	112.689561	39.9949	

Figure 4.6: SQL Result: Lead Time Cancellation Analysis

The results show a strong descriptive relationship between longer lead times and higher cancellation rates in this dataset. Bookings made more than 180 days before arrival have a cancellation rate of approximately 40%.

4.7 Repeat Guest Analysis

```
71  #REPEAT vs NEW GUEST
72  •  select IS_REPEATED_GUEST, count(*) as bookings,
73         sum(FINAL_REVENUE) as total_revenue,
74         avg(ADR) as avg_adr,
75         sum(BOOKING_STATUS='cancelled')*100/count(*)as cancellatio_rate
76  from hospitality_revenue_sql
77  group by IS_REPEATED_GUEST
78  order by total_revenue desc;
```

	IS_REPEATED_GUEST	bookings	total_revenue	avg_adr	cancellatio_rate
▶	0	83952	30534474.65	118.807712	28.7295
	1	3414	475381.06	70.800726	15.3779

Figure 4.7: SQL Result: Repeat VS Non Repeat Guest Analysis

Repeat guests represent a smaller share of the total booking population but have a substantially lower cancellation rate than non-repeat guests.

4.8 Stay Duration Analysis

```

80  #STAY DURATION ANALYSIS
81  •  select TOTAL_NIGHTS, count(*) as bookings,
82         sum(FINAL_REVENUE) as total_revenue,
83         avg(ADR) as avg_adr,
84         sum(BOOKING_STATUS='cancelled')*100/count(*)as cancellation_rate
85  from hospitality_revenue_sql
86  group by TOTAL_NIGHTS
87  order by TOTAL_NIGHTS asc;

```

	TOTAL_NIGHTS	bookings	total_revenue	avg_adr	cancellation_rate
▶	0	651	0.00	0.000000	100.0000
	1	17163	1484348.97	105.781453	18.4117
	2	15742	3009497.34	116.812368	26.7120
	3	17837	5342206.72	122.048492	29.9826
	4	13573	5334050.16	120.147357	30.5017
	5	6741	3480822.04	126.177850	30.3664
	6	3271	2155319.04	134.530309	34.4849

Figure 4.8: SQL Result: Stay Duration Analysis

Stay-duration analysis was performed by grouping bookings according to `TOTAL_NIGHTS`. The dataset contains a number of long-stay records, including a 69-night booking. For this record, the ADR-based room value would be INR 9,108 ($69 \times \text{INR } 132$), while the stored final revenue is INR 6,072. This illustrates why the report uses the validated `FINAL_REVENUE` field for revenue reporting rather than assuming that ADR multiplied by nights is always equivalent to final realized revenue.

4.9 Discount Impact Analysis

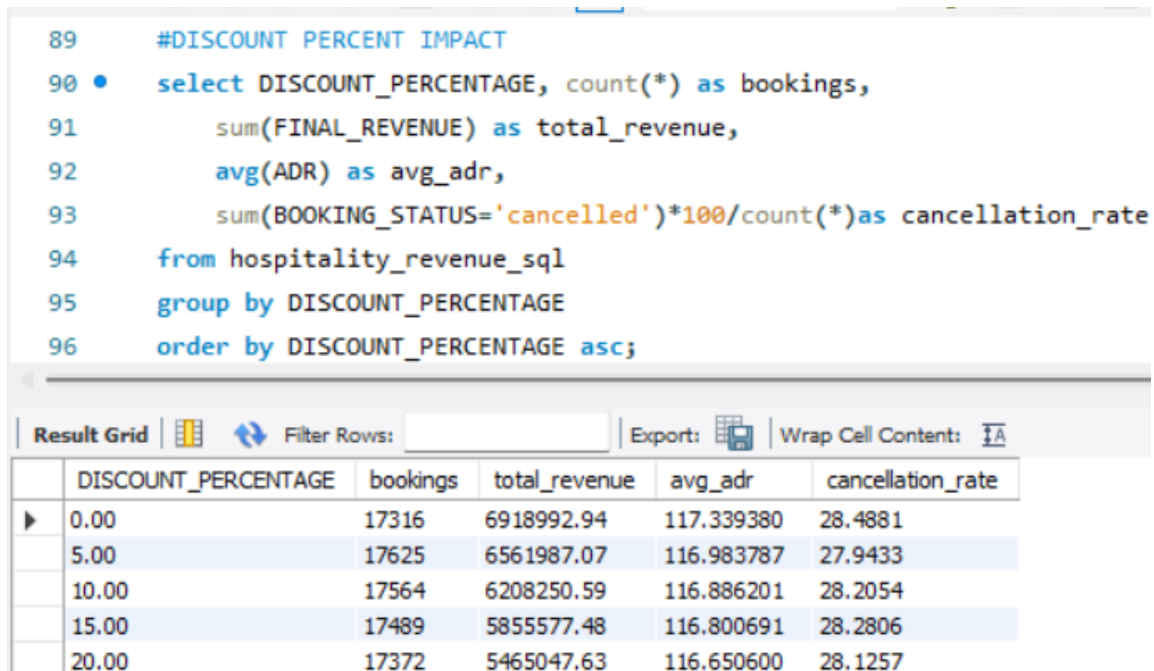


Figure 4.9: SQL Result: Revenue w.r.t Discount

Final revenue decreases progressively across the discount levels in this dataset, while cancellation rates remain relatively close to 28%.

4.10 Customer Segment and Booking Platform Analysis

```

98  #CUSTOMER SEG + BOOKING PLATFORM ANALYSIS
99  •  select CUSTOMER_SEGMENT, BOOKING_PLATFORM, count(*) as bookings,
100      sum(FINAL_REVENUE) as total_revenue,
101      avg(ADR) as avg_adr,
102      sum(BOOKING_STATUS='cancelled')*100/count(*)as cancellation_rate
103  from hospitality_revenue_sql
104  group by CUSTOMER_SEGMENT, BOOKING_PLATFORM
105  order by total_revenue asc;

```

	CUSTOMER_SEGMENT	BOOKING_PLATFORM	bookings	total_revenue	avg_adr	cancellation_rate
▶	Group	Partner Platform	1802	664954.92	118.821032	29.5228
	Group	Website	2643	919526.10	115.365062	27.9228
	Group	Mobile App	4340	1539101.02	117.217486	28.8249
	Individual	Partner Platform	5413	1889775.11	115.115040	28.6902
	Individual	Website	7676	2743123.27	116.956197	28.2439
	Business	Partner Platform	11146	3971977.50	117.034087	28.1536
	Individual	Mobile App	13159	4612695.78	116.484245	27.7225
	Business	Website	15178	5417038.29	116.954161	28.4886
	Business	Mobile App	26009	9251663.72	117.452713	28.0364

Figure 4.10: SQL Result: Customer Segment and Booking Platform Performance

The combined analysis grouped records by both customer segment and booking platform. Business customers using the Mobile App formed the largest individual combination in the result, with 26,009 bookings and approximately INR 9.25M in final revenue. This combined view helps identify high-volume customer-channel combinations rather than examining each dimension separately.

Chapter 5

Tableau Dashboard Development

5.1 Dashboard 1: Revenue Performance Overview

The first dashboard focuses on the overall performance of the hospitality business. The main components include:

- Total Revenue KPI
- Total Bookings KPI
- Average ADR KPI
- Cancellation Rate KPI
- Monthly Revenue Trend
- ADR Trend
- Revenue by Payment Method
- Revenue by Hotel Type
- Booking Status distribution
- Revenue by Booking Platform
- Revenue by Customer Segment

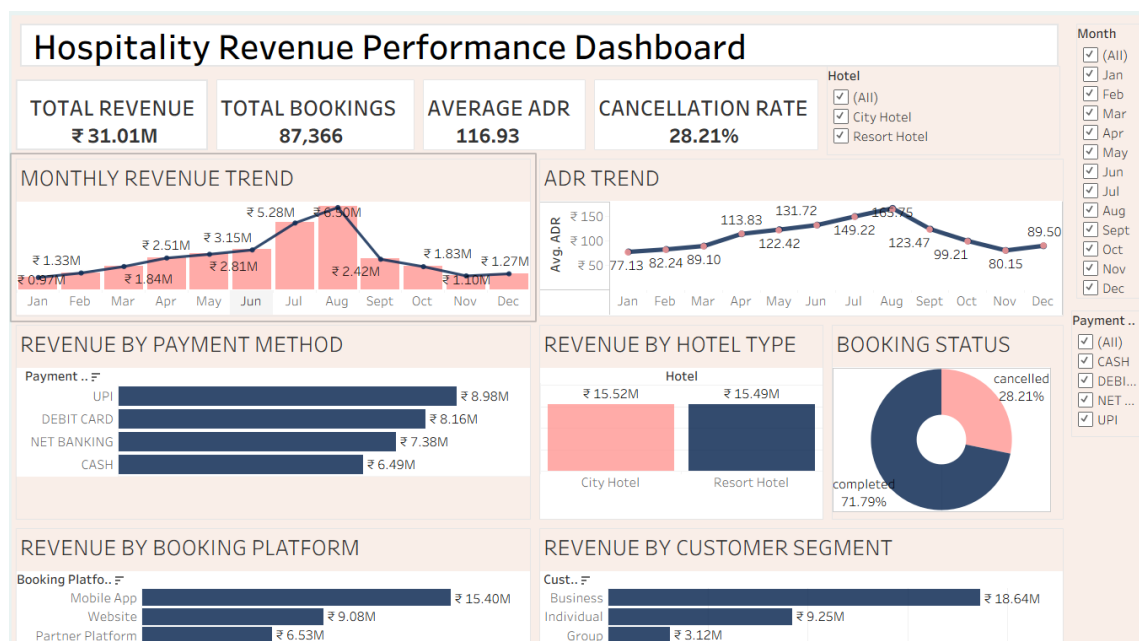


Figure 5.1: Revenue Performance Overview Dashboard

The dashboard provides an interactive overview of key hospitality performance indicators and enables analysis of revenue, bookings, cancellations, customer segments, payment methods, and booking platforms.

5.2 Dashboard 2: Hotel and Customer Insights

The second dashboard focuses on comparisons between hotel types and customer segments. It contains revenue by hotel type, revenue by customer segment, ADR by hotel type and cancellation rate by hotel type.

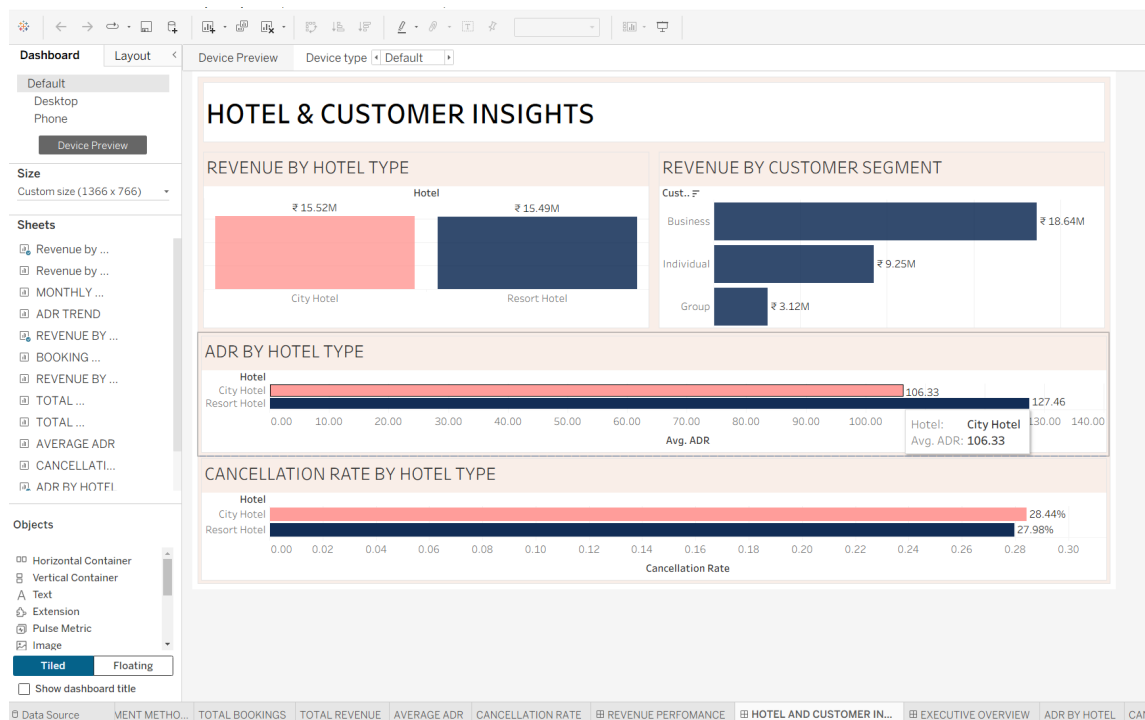


Figure 5.2: Hotel and Customer Insights Dashboard

The dashboard enables comparison of hotel performance and customer segments using revenue, ADR, and cancellation metrics.

5.3 Dashboard 3: Executive Overview

The executive dashboard summarizes the major KPIs and high-level trends using total bookings, total revenue, average ADR, cancellation rate, monthly revenue trend, revenue by payment method and revenue by booking platform.

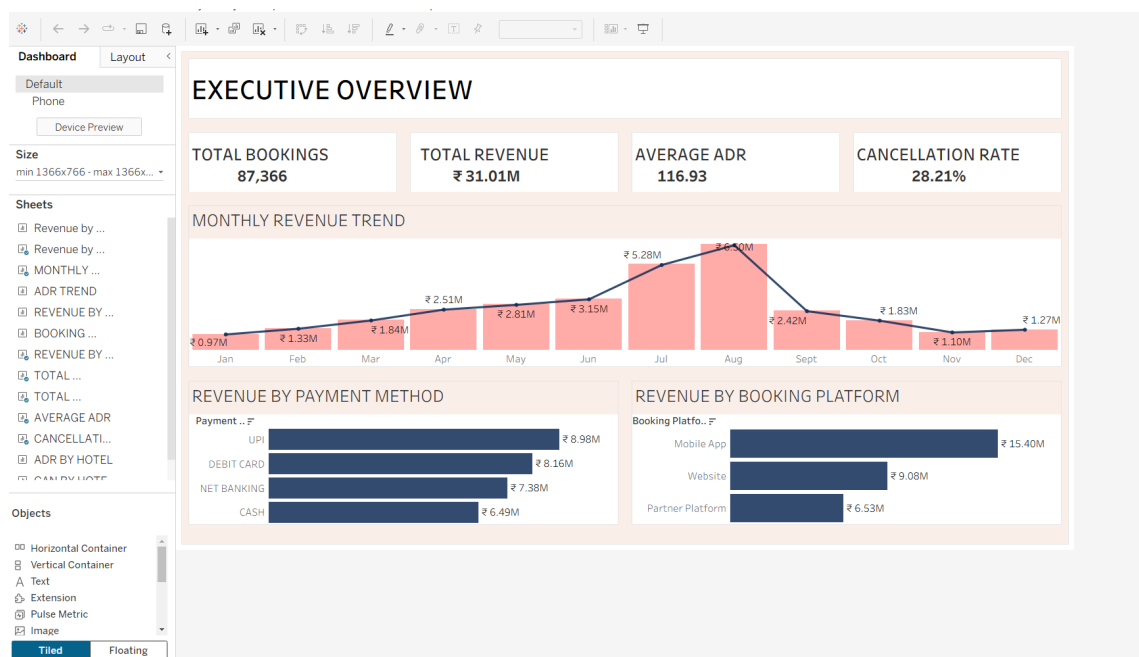


Figure 5.3: Executive Overview Dashboard

The dashboard provides a concise view of the key business metrics and trends required for executive-level analysis.

5.4 Dashboard Design Principles

The dashboards were designed to provide a simple flow from high-level KPIs to detailed business dimensions. Filters were used to allow users to explore the data by relevant dimensions such as hotel, month and payment method. Dashboard actions were reviewed to avoid unintended cross-filtering between unrelated visualizations.

Chapter 6

Key Business Insights

6.1 Insight 1: Overall Revenue and Cancellation

The dataset contains 87,366 bookings and approximately INR 31.01 million in final revenue. The overall cancellation rate is approximately 28.21%, indicating that cancellations represent a significant component of booking activity.

6.2 Insight 2: Business Customers are a Major Revenue Segment

Business customers generated approximately INR 18.64 million from 52,333 bookings. This makes the Business segment the largest contributor to booking volume and final revenue in the analysed dataset.

6.3 Insight 3: Mobile App is a Major Booking Channel

The Mobile App generated approximately INR 15.40 million from 43,508 bookings. It therefore represents the largest booking platform in both volume and final revenue in this dataset.

6.4 Insight 4: Long Lead Times are Associated with Higher Cancellations

Cancellation rates increase from approximately 10.56% for bookings made 0–7 days in advance to approximately 39.99% for bookings made more than 180 days in advance. This is a descriptive association within the historical dataset and does not by itself establish causation.

6.5 Insight 5: Repeat Guests Have Lower Cancellation Rates

Repeat guests show a cancellation rate of approximately 15.38%, compared with approximately 28.73% for non-repeat guests. This indicates a substantial difference in cancellation behaviour between the two groups.

6.6 Insight 6: Higher Discounts Correspond with Lower Final Revenue

Final revenue declines from approximately INR 6.92 million at 0% discount to approximately INR 5.47 million at 20% discount. Cancellation rates remain relatively stable across the discount groups. This pattern should be interpreted as an association in the dataset rather than proof that discounts alone caused the revenue difference.

Chapter 7

Business Recommendations

Based on the descriptive findings, the following areas can be considered for business evaluation:

1. **Monitor long-lead bookings:** Since long-lead bookings have higher observed cancellation rates, management can evaluate confirmation, reminder and cancellation-policy strategies for these bookings.
2. **Strengthen customer retention analysis:** Repeat guests show lower observed cancellation rates, so loyalty and repeat-booking behaviour can be studied further.
3. **Monitor mobile booking performance:** The Mobile App is a major source of booking volume and revenue, making its performance an important KPI for digital distribution analysis.
4. **Evaluate discount effectiveness:** Since higher discount levels are associated with lower final revenue without a large change in cancellation rate, discount campaigns should be evaluated using incremental revenue and profitability rather than booking count alone.
5. **Track hotel-level ADR and revenue together:** Hotel types have similar booking volumes but different ADR levels, so both volume and rate should be monitored when evaluating performance.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

This project demonstrates an end-to-end hospitality analytics workflow using Python, SQL and Tableau. The dataset was inspected and validated before performing business analysis. SQL aggregation was then used to evaluate revenue, bookings, ADR, cancellations, customer segments, booking platforms, lead times, repeat guests, stay duration and discounts. Tableau dashboards transformed these results into an interactive business intelligence interface.

The analysis produced measurable insights about revenue concentration, customer behaviour and cancellation patterns. In particular, business customers and mobile bookings represent major contributors to activity, while long lead times are associated with higher cancellation rates and repeat guests show lower cancellation rates. The project therefore demonstrates how structured data analysis can support evidence-based monitoring of hospitality business performance.

8.2 Future Scope

Future improvements could extend the project from descriptive analytics to predictive and prescriptive analytics. Possible extensions include:

- Predicting booking cancellations using machine learning.
- Forecasting hotel revenue and occupancy.
- Customer lifetime value analysis.
- Dynamic pricing analysis based on demand patterns.

- Cohort analysis for repeat customers.
- Automated refresh of the Tableau dashboard.
- Integration of real-time booking data for operational monitoring.

Appendix A

Selected SQL Queries

A.1 Data Quality: Total Nights Validation

```
SELECT
    COUNT(*) AS total_rows,
    SUM(
        TOTAL_NIGHTS <>
        (STAYS_IN_WEEKEND_NIGHTS + STAYS_IN_WEEK_NIGHTS)
    ) AS incorrect_total_nights
FROM hospitality_revenue_sql;
```

A.2 Discount Impact

```
SELECT
    DISCOUNT_PERCENTAGE,
    COUNT(*) AS bookings,
    SUM('FINAL REVENUE') AS total_revenue,
    AVG(ADR) AS avg_adr,
    SUM(BOOKING_STATUS = 'cancelled') * 100.0 / COUNT(*)
    AS cancellation_rate
FROM hospitality_revenue_sql
GROUP BY DISCOUNT_PERCENTAGE
ORDER BY DISCOUNT_PERCENTAGE;
```

A.3 Customer Segment and Booking Platform

```
SELECT
    CUSTOMER_SEGMENT,
    BOOKING_PLATFORM,
    COUNT(*) AS bookings,
    SUM('FINAL REVENUE') AS total_revenue,
    AVG(ADR) AS avg_adr,
    SUM(BOOKING_STATUS = 'cancelled') * 100.0 / COUNT(*)
        AS cancellation_rate
FROM hospitality_revenue_sql
GROUP BY CUSTOMER_SEGMENT, BOOKING_PLATFORM
ORDER BY total_revenue DESC;
```